

Wetland–Flood Event Model: Variable Construction Reference

“Mark’s Model” pipeline — how every variable is built, and where to improve it

MMSD Wetland–Flood Effectiveness Study

July 2026

This document (i) summarizes the seven pipeline steps and (ii) documents *exactly how each variable is constructed*, with a flagged improvement lever for each — the intended starting point for the variable–refinement pass.

1 The seven steps (at a glance)

#	Step	One line	Script
1	Event extraction	15-min flow $\rightarrow$ Eckhardt baseflow $\rightarrow$ peak picking $\rightarrow$ per-event Y	<code>build_events_flow.py</code>
2	Event rainfall	Stage IV radar QPE, basin-averaged $\rightarrow P_e, V_e$, runoff coeff., API	<code>build_events_rain.py</code>
3	Wetland W, S_w	NWI polygons $\rightarrow \sum A C M f(T)$; Volume–Area storage	<code>build_wetland_nwi.py</code>
4	Controls	drainage area, slope, impervious, K_{sat} /AWS/sand	<code>build_controls.py</code>
5	Panel assembly	join to (gauge $\times$ event); derived terms; collinearity	<code>build_panel.py</code>
6	Models 1–4	mixed-effects / clustered SE; standardize; VIF/placebo/LOGO	<code>fit_models.py</code>
7	Wild cluster bootstrap	honest 9-cluster inference (512 exact sign vectors)	<code>wild_bootstrap.py</code>

2 A. Outcome variables Y (Step 1)

All Y are built on the hourly hydrograph; they inherit the event-segmentation machinery below.

Event segmentation (prerequisite for every Y). **Constructed:** 15-min discharge (USGS 00060) $\rightarrow$ hourly mean. Baseflow via the Eckhardt recursive filter $b_t = [(1 - \text{BFI}_{\text{max}})a b_{t-1} + (1 - a)\text{BFI}_{\text{max}}Q_t] / (1 - a \text{BFI}_{\text{max}})$ with $a = 0.98^{1/24}$, $\text{BFI}_{\text{max}} = 0.60$; quickflow = $\max(Q - b, 0)$. Peaks via `find_peaks` (prominence = $\max(0.02 q_{\text{max}}, P_{95})$, min separation 24 h); event window at 90% of peak prominence; events with >20% interpolated gap dropped. **Improve:** The four knobs { BFI_{max} , prominence, 24h separation, 90% window} define “what counts as a storm” — run a sensitivity analysis; optionally compare an alternative base-flow method (UKIH/sliding-minimum).

Qp_cfs — peak discharge (primary). **Constructed:** window $\max(Q)$. **Improve:** clean; leave as is.

peak_stage_ft — peak stage. **Constructed:** window $\max(\text{gage height})$. **Improve:** only 2017+ (two gauges 2007+); using it as an outcome shrinks the sample; could back-fill from a rating curve.

time_to_peak_hr — time to peak. **Constructed:** $t_{\text{peak}} - t_{\text{start}}$ (rise-to-peak, *internal* to the hydrograph). **Improve:** *highest-value fix*: redefine as $t_{\text{peak}} - t_{\text{rain centroid}}$ (true basin lag) once rainfall timing is joined.

hydro_width_hr — **hydrograph width**. **Constructed:** full width at half prominence (FWHM) of quickflow. **Improve:** switch to duration-above-threshold or multi-height widths; FWHM under-measures multi-peak events.

recession_k_per_hr, recession_time_hr. **Constructed:** slope of $\ln Q$ (peak $\rightarrow$ end) and $t_{\text{end}} - t_{\text{peak}}$. **Improve:** single-exponential linear fit; fit only the stable tail or a two-stage recession.

rb_flashiness — **Richards–Baker**. **Constructed:** $\sum |\Delta Q| / \sum Q$ over the event window. **Improve:** literature R–B is annual/whole-period; ours is event-window — redefine to match if comparing to literature.

quick_vol_m3, total_vol_m3 — **runoff volume**. **Constructed:** $\sum (\text{quick or } Q) \times 3600 \times 0.0283$ (cfs $\rightarrow$ m³s⁻¹ $\times$ hr). **Improve:** quick_vol depends on the baseflow parameters above.

dur_above_q99_hr — **threshold exposure**. **Constructed:** hours with $Q >$ that gauge’s own 99th pct flow. **Improve:** the threshold is a *placeholder*; replace with the NWS flood/bankfull flow or an Atlas-14 return-period flow.

3 B. Rainfall / event forcing (Step 2)

P_e_mm — **event rainfall depth**. **Constructed:** IEM Stage IV daily_precip_in ($0.125^\circ \approx 12$ km, daily). Basin average: sample points at 0.02° inside the catchment $\rightarrow$ snap to the 0.125° lattice $\rightarrow$ weight by sample count per node; sum daily rain over $[t_{\text{start}}, t_{\text{end}}]$ ($\times 25.4$ in $\rightarrow$ mm). **Improve:** *top priority:* (a) upgrade to native 4 km Stage IV or 1 km MRMS (12 km smears convective cells over < 50 km² basins, 1 pixel each); (b) use hourly rather than daily (removes the ~ 5 h UTC/local-day offset); (c) narrow the window to “ N h before peak” so multi-day windows on large rivers don’t merge storms.

V_e_m3 — **rainfall volume**. **Constructed:** $P_e / 1000 \times$ catchment area. **Improve:** follows P_e .

runoff_coeff — **event runoff coefficient**. **Constructed:** quick_vol_m3/ V_e . **Improve:** 1–8% of events exceed 1 (snowmelt / rain-on-snow / rain underestimate) — add a snow model or screen winter; recompute after the rainfall upgrade.

api_30_mm — **antecedent precip index**. **Constructed:** $\sum_{n=1}^{30} 0.9^n P(\text{start} - 1 - n)$. **Improve:** the decay $k = 0.9$ and 30-day window are defaults — calibrate; or use an SSURGO soil-moisture / prior-baseflow proxy.

max_daily_mm — **intensity**. **Constructed:** max daily rain in the window. **Improve:** for true intensity move to hourly intensity + NOAA Atlas-14 return period (not yet computed).

storm_resp_ratio = Q_p / P_e , **peakstage_resp** = stage / P_e . **Constructed:** ratios. **Improve:** follow their parts.

4 C. Wetland effectiveness W and storage S_w (Step 3) — main improvement zone

A_i — **wetland area**. **Constructed:** USFWS NWI polygon area (EPSG:32616), tiled fetch, clipped to catchment. **Improve:** merge/replace with the finer Wisconsin Wetland Inventory; optionally filter by type (drop open water).

T_ij — **travel time wetland $\rightarrow$ gauge**. **Constructed:** sampled at the wetland *centroid* from the routing surface T_j (10 m); NaN/ ≤ 0 fall back to τ . **Improve:** *core issue:* absolute travel time is

uncalibrated (SLOPE_MIN, Manning n , channel celerity are defaults) $\Rightarrow$ only a relative index. Calibrate celerity from observed lag/tracers so a fixed τ becomes physical; sample polygon-minimum T (nearest exit) instead of centroid.

- C_i — connectivity. Constructed:** near-channel: 1 if distance to the network ($\text{acc} > 0.5 \text{ km}^2$) $< 100 \text{ m}$ (C_bin); graded variant $e^{-d/100}$; distance via Euclidean transform, sampled at centroid. **Improve:** adopt the EPA three-class connectivity (surface / shallow-GW / deep-GW); sensitivity on the 100 m buffer; use nearest polygon edge.
- M_i — Cowardin type modifier. Constructed:** from NWI ATTRIBUTE: emergent 1.0; forested/scrub-shrub 0.8; open-water/riverine 0.5; lacustrine 0.3; else 0.7. **Improve:** this is a *judgment table* — best upgrade: use HGM (hydrogeomorphic) class (floodplain vs. isolated depression have measured peak-attenuation differences) or literature weights; optionally layer in water regime.
- f(T), τ — decay kernel. Constructed:** $f_{\text{exp}} = e^{-T/\tau}$, $f_{\text{harm}} = 1/(1+T/\tau)$, τ = catchment-median T (adaptive, because T is uncalibrated). **Improve:** after calibrating T , switch to a physical τ (connectivity literature) and model-compare exp vs. harmonic.
- W_j — cumulative effectiveness (treatment). Constructed:** $\sum_i A_i C_i M_i f(T_{ij})$; six stored variants (W_exp primary, W_harm , W_exp_noC , W_exp_noM , W_exp_grad , W_area). Modeling uses the intensive $W_n = W_{\text{exp}}/\text{area}$. **Improve:** each multiplier (A, C, M, f) is independently upgradable; W_{exp} is extensive and collinear with DA ($r=0.89$), hence normalized — try alternative normalizations (per channel length, per upstream area).
- Sw_va_m3 — storage capacity. Constructed:** Volume–Area power law $V_i = 0.05 A_i^{1.2}$, $S_w = \sum V_i$; a DEM depression-volume cross-check is uninformative here. **Improve:** constants $c=0.05, \beta=1.2$ are literature-range defaults (order of magnitude only) — fit a Wisconsin-specific V – A , or use raw-LiDAR depression-fill volume (*not* the routing-conditioned DEM, which was built to remove depressions); consider type-specific c, β .

5 D. Control variables (Step 4)

- DA_km2 — drainage area. Constructed:** catchment polygon area (validated $\pm 5\%$ vs. NWIS). **Improve:** clean.
- basin_slope, chan_slope. Constructed:** $\sqrt{g_x^2 + g_y^2}$ on the 3DEP 10 m DEM, averaged over all catchment cells / over channel cells ($\text{acc} > 5000$). **Improve:** channel slope could be the main-stem longitudinal profile; add basin relief, drainage density.
- imp_pct — impervious fraction. Constructed:** NLCD impervious (nlcd_bygeom) for 2001/2011/2021, basin mean, linearly interpolated to each event's year. **Improve:** move to Annual NLCD for a true urbanization trajectory; add engineered storage (CSO/storm-sewer, detention, Deep Tunnel) — currently unmodeled.
- ksat_log — infiltration. Constructed:** POLARIS ksat_0_5cm basin mean ($\log_{10} \text{ cm h}^{-1}$). **Improve:** surface layer only; add deeper layers, or use SSURGO Hydrologic Soil Group (A–D), which the workflow explicitly requests.
- sand_pct, aws_mm. Constructed:** POLARIS sand_0_5cm ; gNATSGO aws0_100 , basin means. **Improve:** collinear with K_{sat} and the urban–rural gradient — keep one in the model.

6 E. Derived modeling terms (Steps 5–6)

Wn = $W_{\text{exp}} / (\text{DA} \times 10^6)$ — intensive treatment (breaks W – DA collinearity).

sat = $\log(V_e/S_w)$ — saturation ratio. **WP** = $z(W) \cdot z(P)$, **WS** = $z(W) \cdot z(\text{sat})$ — interactions.

Standardization all predictors z -scored; log taken for $Q_p, P_e, V_e, \text{DA}, S_w, W_{\text{exp}}$.

usable = $(P_e > 1) \wedge (\text{runoff_coeff finite})$. **Improve:** the interaction terms are currently the *only* identified signal (within-gauge variation) — prioritize the quality of W, P , and sat that feed them.

Priority order for the refinement pass

1. P_e — native 4 km / 1 km hourly radar (biggest effect on every event metric).
2. M_i — replace the judgment table with an HGM classification.
3. T_{ij} / τ — calibrate absolute travel time (celerity) so a physical τ works.
4. S_w — Wisconsin-specific Volume–Area or LiDAR depression volume.
5. **time_to_peak** — redefine as rainfall-centroid-to-peak basin lag.